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Real—world application of lecanemab in early—stage alzheimer's disease: a single—center prospective cohort analysis

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Abstract

Background and objectives Lecanemab (Leqembi®) has been approved for the treatment of early Alzheimer's disease (AD) patients. However, the safety and efficacy of lecanemab in clinical practice in Asia population remain unclear.

Design, setting, and participants This prospective cohort study was conducted in a single center in Eastern China, including 76 early-stage AD participants treated with lecanemab. All participants underwent at least 1 lecanemab infusion.

Results The mean age was 66 years (65.53 ± 9.31), and 51 (67.1%) participants being female. A total of 49 (64.5%) participants were *ApoE-ε4* carriers, including 34 (44.7%) heterozygotes and 15 (19.7%) homozygotes. Infusion-related side effects (IRSE), primarily occurred after the first infusion, were observed in 14 participants (18.4%). The most common IRSE were fever and fatigue. Until 9 Aug 2025, 58 participants had received treatment for more than 3 months, 44 for more than 6 months and 12 for more than 12 months. Amyloid-related imaging abnormalities (ARIA) were observed in 11 participants. Specifically, solitary ARIA-H (hemorrhage) was detected in 7 participants, while ARIA-E (edema) with or without cerebral micro-hemorrhage was identified in 4. Notably, all the 11 participants with ARIA were asymptomatic. Participants with isolated ARIA-H exhibited higher baseline Fazekas scores of PVWMHs ($p = 0.008$). Participants with a Clinical Dementia Rating Scale Global Score (CDR-GS) of 1 had a more rapid decline in Mini-Mental State Global Score (MMSE) scores as compared to those with a CDR-GS of 0.5. Following 12 months of lecanemab therapy, amyloid PET exhibited a significant reduction in brain amyloid burden.

Conclusion These data support that lecanemab appears to be generally tolerated in Asian population. The IRSE and ARIA-E events were less common than the Clarity AD study.

Keywords Alzheimer's disease, Lecanemab, Amyloid-related imaging abnormalities, Cognitive progression

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Introduction

Alzheimer's disease (AD), characterized by progressive cognitive disorder and abnormal behavior, is the most common cause of dementia, imposing a significant social and economic burden [21]. Extracellular amyloid plaques formed by aggregated amyloid-β peptides (Aβ) and intracellular neurofibrillary tangles (NFT) composed of hyperphosphorylated tau are the two main pathological hallmarks of AD. Treatments for the disease in the past are largely limited to symptomatic therapy, including excitatory amino acid receptor antagonists and cholinesterase inhibitor. The approval of anti-amyloid monoclonal antibodies, a disease-modifying therapy, has significantly changed the management of early-stage AD.

Lecanemab (BAN2401) is a humanized IgG1 monoclonal antibody that binds to soluble Aβ aggregates and has high selectivity over monomer (> 1000 fold) and insoluble fibrils (approximately 10 ~ 15 fold) [24, 27]. Lecanemab was fully approved by the United States (US) Food and Drug Administration (FDA) in January 2023 and received approval for market launch in mainland China in January 2024. The drug's efficacy was validated through both biomarker assessments and clinical scales [8]. In the phase 3 Clarity AD trial, lecanemab demonstrated clinical efficacy by slowing the cognitive and functional decline in patients with early-stage AD, accompanied by the clearance of Aβ deposition [28].

The most significant concerns about the anti-Aβ antibodies are the treatment-associated magnetic resonance imaging (MRI) signal abnormalities, which are termed amyloid-related imaging abnormalities with edema/effusion (ARIA-E) or hemorrhage/hemosiderin deposits (ARIA-H). ARIA-E refers to the extravasation of fluid resulting in interstitial vasogenic oedema or sulcal effusion in the leptomeningeal/subpial space [3, 25]. ARIA-H refers to micro-hemorrhages (MCH) or superficial siderosis observed as hypointense hemosiderin deposition [2, 3, 26]. In the Clarity AD trial, the incidence of ARIA-E was 12.6%, while that of ARIA-H was 17.3% [28] throughout the 18-month treatment. Subgroup analysis of the Asian population revealed a reduced incidence of 6.2% and 14.4%, respectively [5]. Most cases with ARIA-E/H are asymptomatic and can be resolved with dose modification or drug suspension or cease in 3 ~ 4 months [6, 7]. Infusion-related side effects (IRSEs) associated with lecanemab occurred in 26.4% of the overall Clarity AD population but declined notably to 12.3% in Asian subgroup [5, 28].

Given the distinct profile of lecanemab observed in Asian populations (primarily enrolled from Japan and South Korea) in the Clarity AD trial, characterized by a favorable risk–benefit ratio and manageable safety risks, it is imperative to validate its application within Chinese clinical practice. As Asia's inaugural real-world evidence initiative, this prospective cohort study, aims to delineate lecanemab's disease-modifying effects in routine clinical practice settings by dynamically tracking longitudinal trajectories of clinical and neuroimaging outcomes.

Methods

Participant selection and treatment

This is an 18-month, single-center, hospital-based, longitudinal, prospective observational study involving patients with early AD (ClinicalTrials.gov ID: NCT06741553; date of registration: 2024–11–29). A total of 76 early AD participants were prospectively enrolled from the memory clinic at the Second Affiliated Hospital of Zhejiang University between 28 Jun 2024 and 9 Aug 2025. Fifty-eight participants had received treatment for more than 3 months, 44 for more than 6 months and 12 for more than 12 months. Two participants discontinued treatment after 3 and 8.5 months, respectively, due to family-related circumstances. One participant, who had fever during the 1st infusion, discontinued treatment at the family's request owning to concerns about IRSEs. Early AD was defined as mild cognitive impairment (MCI) or mild dementia due to Clinical Dementia Rating Scale Global Score (CDR-GS) between 0.5 to 1, supported by objectively defined amyloid positivity on positron emission tomography (Aβ-PET). The exclusion criteria consist of (1) stroke, transient ischemic attack, bleeding disorders, or seizures in the previous 12 months, (2) warfarin, vitamin K antagonists, direct oral anticoagulants, heparin, acute thrombolytics, and clotting disorders, (3) intolerant of MR scanning, (4) with MRI evidence of hemorrhagic manifestation (over 4 microhemorrhages which are 10 mm or less at the greatest diameter or a single macro-hemorrhage which is greater than 10 mm at the greatest diameter), (5) with unstable psychiatric disorder. All participants provided written informed consent approved by the Ethics Committee. Consent encompassed sharing of de-identified demographic, genetic, laboratory, imaging, and clinical data. The demographic data of participants were collected through inquiry.

Lecanemab was administered intravenously every other week. Lecanemab was added to 0.9% sodium chloride. Dosing was weight-adjusted, and participants were administered 10 mg per kilogram of body weight according to the Appropriate Use Recommendations (AURs) [6]. The infusion lasted about 1 h and the participants were required to stay three hours for observation in case of the occurrence of infusion reactions.

Apolipoprotein E genotyping

All participants received Apolipoprotein E (*ApoE*) genotyping and were subsequently counseled on their *ApoE* status and relative risk of ARIA by the treating physician.

MRI neuroimaging

Non-contrast brain MRI were scanned by using Shanghai United-imaging uMR790-3.0 T with sequences including T1, T2 fluid attenuated inversion recovery (FLAIR, 5 mm slice thickness), susceptibility weighted imaging (SWI, 1 mm slice thickness) and diffusion weighted imaging (DWI). Acute cerebral infarction was detected by DWI sequence. The total MRI scan time was 23 min and 5 s. Brain MRIs were obtained at baseline, and 2, 3, 6, 12, 18 months after the 1st infusion as AURs recommended. All images were blindly evaluated by two experienced neuroradiologists without access to clinical information. Periventricular and deep white matter hyperintensities were assessed using the Fazekas scale: periventricular white matter hyperintensities (PVWMH) were defined as follows: grade 0 (no lesions), grade 1 (cap-like or pencil-thin lining lesions), grade 2 (smooth halo-like lesions) and grade 3 (Irregular PVWMH extending into deep white matter); deep white matter hyperintensities (DWMH) were defined as follows: grade 0 (no lesions), grade 1 (punctate foci), grade 2 (beginning confluent foci) and grade 3 (large confluent lesions). Hippocampal atrophy was evaluated using the Medial Temporal Atrophy (MTA) scale: grade 0 (no atrophy), grade 1 (widening of the choroidal fissure only), grade 2 (additional enlargement of the temporal horn of the lateral ventricle), grade 3 (moderate volume loss of the hippocampus) and grade 4 (severe volume loss of the hippocampus) [19]. The mean grade of both sides was calculated for analysis. Cerebral Micro-hemorrhages (MCH) were counted with SWI. Participants were divided into two groups (with MCH or without) according to their baseline SWI.

Suspected ARIA were estimated and graded using T2 FLAIR (ARIA-E) and SWI (ARIA-H) sequences by both the on-site radiologist and the treating physician. The severity of ARIA-E was defined as follows: mild (single FLAIR hyperintensity confined to sulcus and/or cortical/subcortical white matter in one location < 5 cm), moderate (single FLAIR hyperintensity of 5–10 cm, or involvement of > 1 site, each measuring < 10 cm), or severe (any FLAIR hyperintensity measuring > 10 cm) [9, 25]. The severity of ARIA-H was defined according to the number of new incident MCHs (mild < 4, moderate 5–9, severe > 10) or number of new focal areas of superficial siderosis (mild = 1, moderate = 2, severe > 2) [9, 25]. Dosing continues for individuals with radiographically mild asymptomatic ARIA recognized only on MRI with careful monitoring for symptoms and monthly non-contrast MRI scans until ARIA-E has resolved and ARIA-H has stabilized. For those with moderate or severe radiographic ARIA or those with symptoms, the lecanemab was discontinued. Clinical symptoms were carefully monitored and monthly non-contrast MRI scans were conducted. After ARIA-E has resolved and ARIA-H has stabilized, whether to restart lecanemab was comprehensively discussed with their families considering the risks and benefits.

Clinical assessments

All participants received comprehensive cognitive assessments conducted by medical specialists before the 1st infusion and 3, 6, 12, 18 months after the 1st infusion. The assessments used are as follows: Mini Mental State Examination (MMSE), CDR-SB, Alzheimer's Disease Cooperative Study-Activity of Daily Life (ADCs-ADL) and Neuropsychiatric Inventory (NPI). Participants who suspended medication because of moderate or severe ARIA were also assessed routinely to observe their cognitive changes.

Infusion-related side effects

Given the potential occurrence of infusion-related side effects (IRSE), patients should be monitored for 3 h following the initial infusion, with a subsequent telephone follow-up conducted on the same day, as delayed infusion reactions may occasionally occur [6]. IRSE were managed according to AURs of lecanemab [6].

PET imaging

Participants received PET imaging at baseline and 12 months after initiating lecanemab. All amyloid PET imaging were acquired using Siemens scan model Biograph mCT. Participants received a bolus injection of 10 mCi 18F-Florbetapir (AV-45), and a 10-min static scan was acquired at 50-min post-injection. Mean cortical standardized uptake value ratio (SUVr) was calculated as the average of six regions of interest (ROIs) (medial orbital frontal, lateral temporal, parietal, anterior cingulate, posterior cingulate, and precuneus) with respect to whole cerebellum [16, 31].

Statistics

All statistics were analyzed with SPSS (Version 27, IBM). The significance of differences was assessed with *t* tests for continuous variables conforming to a normal distribution, Mann–Whitney *U* test for continuous variables conforming to a skewed distribution, Chi-square test and Fisher exact tests for ordinal variables. Paired samples *t*-test was employed to compare the MMSE variance before and after ARIA occurrence. Participants were categorized according to baseline CDR-GS (MCI group, CDR-GS = 0.5; mild-dementia group, CDR-GS = 1) and age (early-onset AD [EOAD] group, < 65 y; late-onset AD [LOAD] group, ≥ 65 y). Considering that cultural difference and education can deeply affect MMSE scores, participants in MCI subgroup were categorized according to baseline MMSE scores (< 20 or ≥ 20) to investigate the efficacy of lecanemab in participants with lower MMSE

scores. To investigate longitudinal cognitive progression trends, mixed linear models and generalized mixed linear models were implemented with statistics from baseline to 12-month visit. Fixed effects included group, time, age, education, periventricular and deep white matter hyperintensities (Fazakas grades) and group × time interaction. Random effects included time in both intercept and slope. Paired samples *t*-test was employed to evaluate changes in amyloid burden before versus after lecanemab treatment. Plots were created with Prism version 9.

Results

Demographics

Seventy-six early-stage AD participants were enrolled in this study (Fig. 1). They were treated with 15.2 (SD, 8.2) infusions and had 3.3 (SD, 1.4) monitoring MRI scan in average (Fig. 2). Demographic and baseline clinical characteristics of the participants are exhibited in Table 1 and Supplementary Fig. 1. *ApoE ε4* carriers comprised 64.5% of participants (34 heterozygotes [44.7%]; 15 homozygotes [19.7%]). Educational attainment was categorized as illiterate (n = 6, 7.9%), middle school or lower (n = 41, 53.9%), and post-middle school education (n = 29, 38.2%). The most prevalent comorbidities were hypertension (n = 14), diabetes mellitus (n = 6), hyperlipidemia (*n* = 10), and thyroid disorders (n = 7). Baseline MRI revealed cerebral microhemorrhages (1–4) in 41 participants (53.9%). Periventricular white matter hyperintensities (PVWMHs) were present in 72 participants (94.7%) with Fazekas grade distributions of: grade 1 (n = 41, 53.9%), grade 2 (n = 25, 32.9%), and grade 3 (n = 6, 7.9%). Deep white matter hyperintensities (DWMHs) were observed in 72 participants (94.7%) with grade distributions of: grade 1 (n = 41, 53.9%), grade 2 (n = 23, 30.3%), and grade 3 (n = 8, 10.5%).

Infusion-related side effects (IRSE)

IRSE are shown in Table 2. Infusion reactions predominantly occurred following the first infusion (14 participants, 18.4%) and were mild to moderate in severity. The most common IRSE were fever and fatigue. All IRSE occurred within 24 h post-infusion and could be

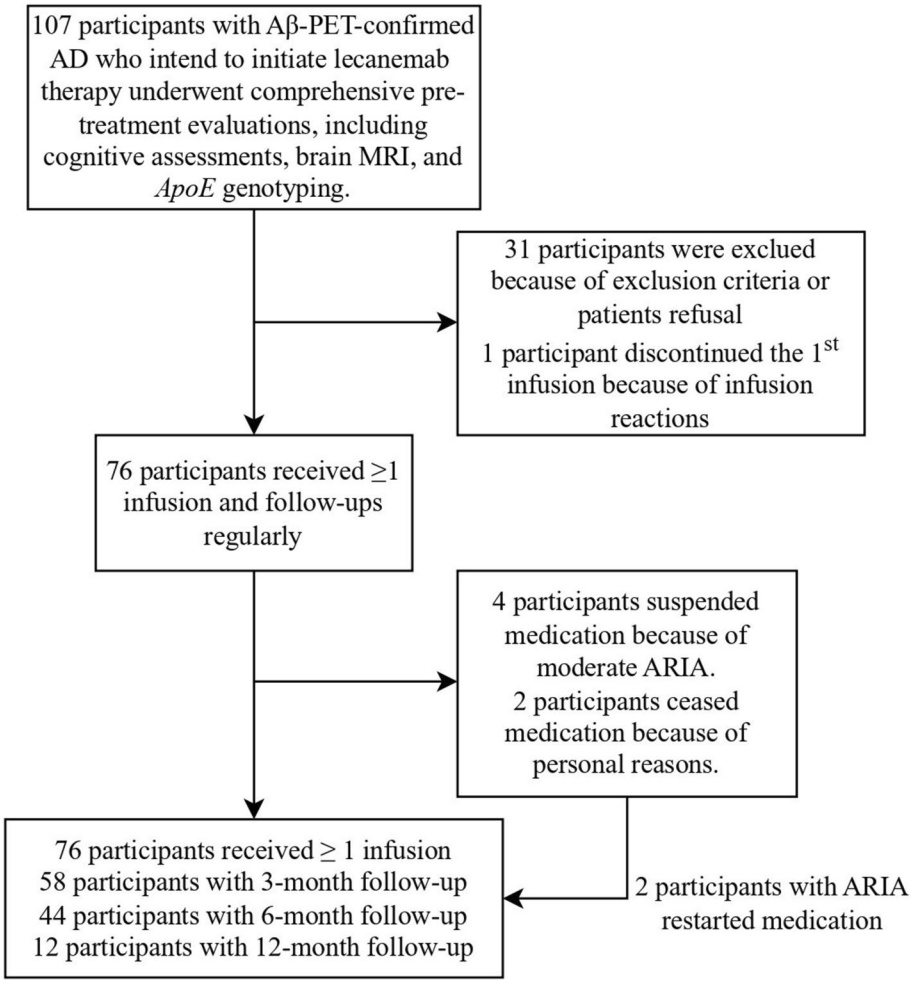

Fig. 1 Algorithm of the process for participants inclusion and follow-up over the 12-month period of this study

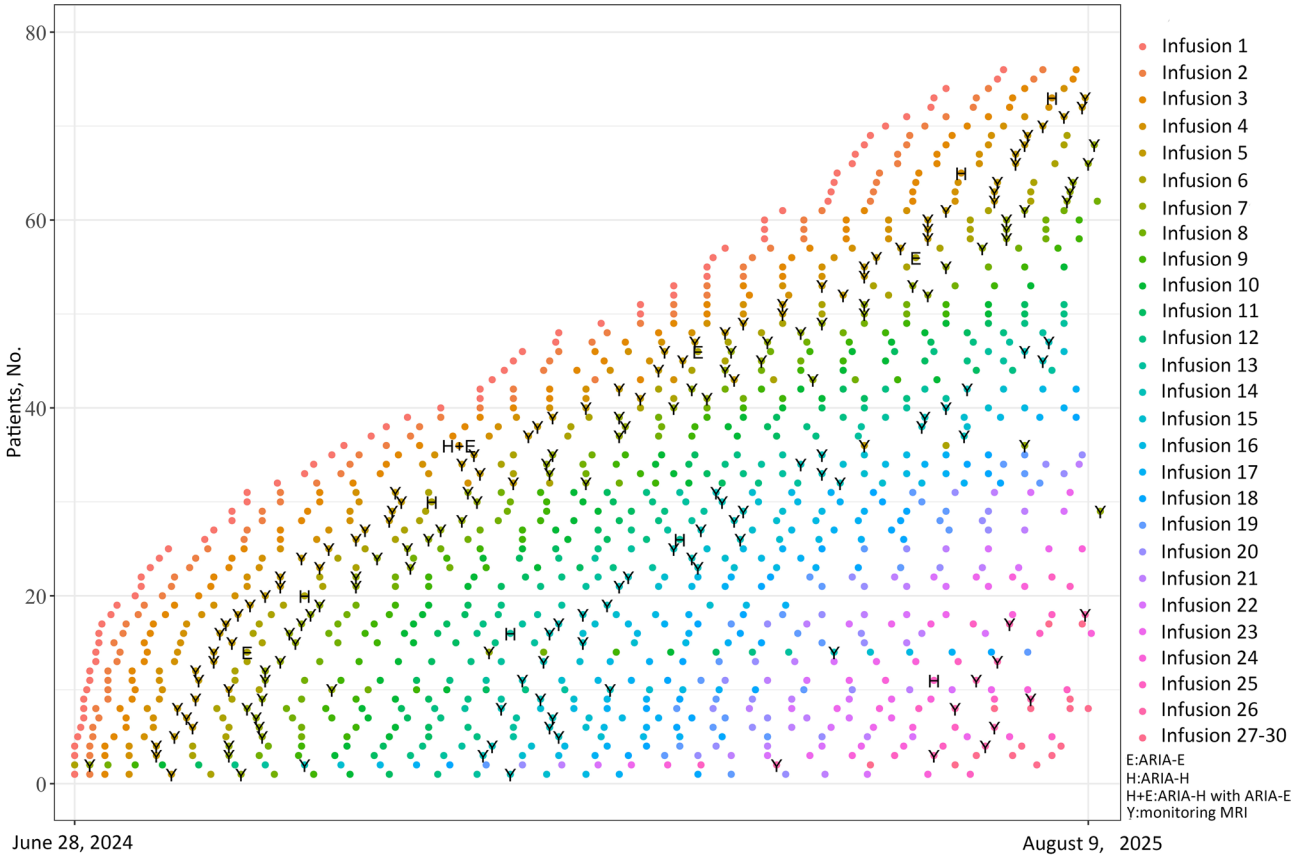

Fig. 2 Lecanemab Infusions, Monitoring Magnetic Resonance Imaging (MRI) Scans, and Amyloid-Related Imaging Abnormalities (ARIA) Over Time. Each circle indicates 1 infusion with the color denoting the number of the infusion. E indicates ARIA-E; H, ARIA-H; H + E, ARIA-E with ARIA-H; Y, monitoring MRI

alleviated with rest or symptomatic interventions. Only participants who experienced high fever or rash following infusion were pretreated with acetaminophen and loratadine/phenergan until they remained asymptomatic after the subsequent infusion. Comparison of IRSE across different cohorts are presented in Table 3.

ARIA

No participants developed symptomatic ARIA that necessitated urgent MRI. All ARIA cases were detected during scheduled surveillance scans. Only participants who received ≥ 4 lecanemab infusions and completed ≥ 1 monitoring MRI were considered at risk for ARIA. Fifty-eight participants underwent monitoring MRI at 2 and 3 months after starting lecanemab infusion, 44 completed the 6-month scan, and 12 completed the 12-month scan (Supplementary Table 1). The precise timing of ARIA onset is detailed in Table 4 and Fig. 2. Among the 11 participants, 4 had ARIA-E with or without ARIA-H and 7 had isolated ARIA-H. All ARIA-H events were confined to micro-hemorrhages. Neither superficial siderosis nor macro-hemorrhages were observed. All cases were asymptomatic. Radiographically, 5 ARIA cases were moderate and 6 were mild in severity. The majority of ARIA events (8/11) occurred within the first three months after the initial infusion (Fig. 3). Detailed characteristics of all ARIA cases are presented in Table 4. No significant cognitive decline was observed when comparing MMSE scores obtained before ARIA-E onset with those reassessed 3–6 months ARIA-E occurrence ($p = 0.128$). Similar findings were observed in ARIA-H cases (Table 4; Supplementary Table 2). Participants with isolated ARIA-H exhibited higher baseline Fazekas scores of PVWMHs ($p = 0.008$, Supplementary Table 3). Table 3 compares the incidence and severity of ARIA within the first 6 months of treatment across different cohorts.

Cognitive assessments

Cognitive assessments were routinely conducted and recorded (Supplementary Table 4). To identify factors influencing longitudinal cognitive change, participants were categorized based on their baseline cognitive state using CDR-GS (MCI group, CDR-GS = 0.5; mild-dementia group, CDR-GS = 1) and age (EOAD, age < 65y; LOAD, age ≥ 65y). Linear mixed models were employed, adjusting for age, education level, and white matter hyperintensities (Fig. 4 and Supplementary Table 5). The MCI group exhibited a less rapid decline over time compared

Table 1 Socio-demographic and clinical characteristics

Characteristics [Mean ± SD/n (%)]	*N* = 76
Age at onset	65.53 ± 9.31
Sex	
Male	25 (32.9%)
Female	51 (67.1%)
ApoE genotype	
ε4 non-carrier	27 (35.5%)
ε4 carrier	49 (64.5%)
heterozygote	34 (44.7%)
homozygote	15 (19.7%)
Education	9.32 ± 4.67
Illiterate	6 (7.9%)
Middle school education or below	41 (53.9%)
Post-middle school education	29 (38.2%)
Anti-platelet agents	9 (11.8%)
Symptomatic medications	
AChEI	32 (42.1%)
Sodium Oligomannate	27 (35.5%)
Memantine	3 (3.9%)
Antidepressant drugs	9 (11.8%)
Comorbidity	
Hypertension	14 (18.4%)
Diabetes mellitus	6 (7.9%)
Hyperlipidemia	10 (13.2%)
thyroid disorder	7 (9.2%)
Depression	2 (2.6%)
Sleep disorder	2 (2.6%)
Cerebral Micro-hemorrhage	
With	41 (53.9%)
Without	35 (46.1%)
Periventricular White matter Fazekas grade	
0	4 (5.3%)
1	41 (53.9%)
2	25 (32.9%)
3	6 (7.9%)
Deep White matter Fazekas grade	
0	4 (5.3%)
1	41 (53.9%)
2	23 (30.3%)
3	8 (10.5%)
Hippocampal MTA grade	
1.0	20 (26.3%)
1.5	3 (3.9%)
2.0	41 (53.9%)
2.5	1 (1.3%)
3.0	10 (13.2%)
3.5	0 (0.0%)
4.0	1 (1.3%)
MMSE	19.55 ± 5.00
CDR-SB*	3.12 ± 1.19
CDR-GS	
0.5	66 (86.8%)
1	10 (13.2%)
ADCs-ADL*	69.29 ± 5.98
NPI*	4.86 ± 6.50

Abbreviations ApoE, Apolipoprotein E, *AChEI* acetylcholinesterase inhibitor, *MMSE* Mini-Mental State Global Score, *CDR-SB* Clinical Dementia Rating-Sum of Boxes, *CDR-GS* Clinical Dementia Rating Scale Global Score, *ADCs-ADL* Alzheimer's Disease Cooperative study-Activity of Daily Life, *NPI* neuropsychiatric inventory

*3 participants had CDR-SB missing, 6 participants had ADCs-ADL scores missing and 4 participants had NPI scores missing

Table 2 Infusion-Related Side Effects (IRSE)

Infusion	No. (%) of participants with IRSE	IRSE	No. (%)
1st	14/76 (18.4%)	Fever	8 (10.5%)
		Rash	1 (1.3%)
		Fatigue	3 (1.3%)
		Headache	1 (1.3%)
		Dizziness	1 (1.3%)
		Edema	2 (2.6%)
2nd	4/76 (5.3%)	Fever	1 (1.3%)
		Rash	1 (1.3%)
		Fatigue	1 (1.3%)
		Edema	1 (1.3%)
3rd	1/76 (1.3%)	Fever	1 (1.3%)
		Dizziness	1 (1.3%)
4th	0/73 (0.0%)	/	/

Abbreviation IRSE infusion-related side effects

to the mild-dementia group (β [95% CI]: 0.537 [0.099, 0.974]; $p = 0.019$). Additionally, MMSE scores in the EOAD group declined more rapidly over time than those in the LOAD group (β [95% CI]: −0.434 [−0.726, −0.141]; $p = 0.004$). Over the 12-month period, no significant differences in changes in ADCs-ADL and CDR-SB were observed between the MCI and mild-dementia groups (Supplementary Table 6). Given the impact of educational level and cultural differences on MMSE scores, we included participants with MCI who were treated with lecanemab, primarily based on their CDR-GS scores. To further assess whether participants with lower MMSE scores experienced more rapid cognitive decline, we analyzed the cognitive progression of participants categorized into subgroups defined by baseline MMSE scores (< 20 and ≥ 20). No significant differences in the rate of MMSE decline were observed between the subgroups over the 12-month period time ($p > 0.05$; Supplementary Table 6).

Table 3 Adverse events across studies

	Current cohort ($n=76$; $n=58$ for ARIA)[a]	Asian Clarity AD ($n=146$)	Overall Clarity AD ($n=898$)	A US MDC cohort[b] ($n=234$; $n=194$ for ARIA)[c]
	No. (%)			
ARIA-E with or without ARIA-H[d]	4 (6.9)	/	104 (11.6)	25(12.9)
Isolated ARIA-H[d]	6 (10.3)	/	33 (3.7)	/
Infusion-related side effects	14 (18.4)	18 (12.3)	237 (26.4)	87 (37.2)
Death	0	0	6 (0.7)	0
Adverse events leading to study drug dose adjustment	7 (9.2)	16 (11.0)	219 (24.4)	/
Drug withdrawal	1 (1.3)	1 (0.7)	64 (7.1)	23(9.8)
Drug suspension	4 (5.3)	15 (10.3)	175 (19.5)	/
Infusion interruption	2 (2.6)	5 (3.4)	22 (2.4)	/

[a]76 participants received at least 1 infusion, and 58 participants who received ≥ 4 lecanemab infusions and completed ≥ 1 monitoring MRI were considered at risk for ARIA

[b]A retrospective cohort recruited at the Washington University Memory Diagnostic Center [18]

[c]234 participants received at least 1 infusion, and 194 participants who received ≥ 4 lecanemab infusions and completed ≥ 1 monitoring MRI were considered at risk for ARIA

[d]To be comparable, only ARIA occurred before 6-month follow-up were considered in the comparison

Pathological changes

Amyloid-PET was completed by 12 participants. Changes in mean amyloid-β PET SUVr were shown in the Fig. 5 and Supplementary Table 7. The brain amyloid burden declined significantly after lecanemab treatment ($p=0.004$; 95% CI [−0.303, −0.076]). One participant exhibited an increased amyloid burden following 12 months of treatment (SUVr from 1.278 to 1.523).

Discussion

The primary objective of this study was to evaluate the feasibility and safety of lecanemab treatment, as well as the cognition development of lecanemab-treated participants, in an Asian cohort recruited within a real-world clinical practice setting. This study enrolled 76 early-stage AD participants who received at least one dose of lecanemab. Among them, 58 completed at least one MRI follow-up, with 11 cases developing radiologically confirmed asymptomatic ARIA within the 12-month treatment. The study included: (1) monitoring infusion-related adverse events; (2) investigating risk factors for ARIA occurrence; (3) comparing MMSE scores before and after ARIA onset; (4) describing cognitive progression in lecanemab-treated AD participants categorized by baseline cognitive level and age and (5) detecting amyloid burden change after medication.

Lecanemab was generally tolerated in this Asian cohort. The incidence of IRSEs (18.4%) tended to be lower than that reported in the overall Clarity AD cohort (26.4%) [28], but higher than that in the Asian regional cohort (enrolled in Japan, South Korea and Singapore) from Clarity AD (12.3%) [5]. The majority of IRSEs (73.7%) occurred after the first infusion and were predominantly mild in severity, resolving with rest. Common symptoms included fever, fatigue, rash, and headache. Prophylactic medication was not routinely administered. However, participants who experienced high fever or rash requiring symptomatic treatment following infusion were premedicated with acetaminophen and loratadine/phenergan until they remained asymptomatic after the subsequent infusion. No participants experienced infusion reactions exceeding grade 2, as defined by the AURs for lecanemab [6]. These findings suggest that the Asian population exhibits well tolerance to lecanemab, experiencing lower frequencies of IRSEs with milder symptoms. Consequently, in Asian populations, prophylactic medication could be reserved for personalized administration to higher-risk individuals. Additionally, close monitoring within the first 24 h after infusions remains warranted.

ARIA-E predominantly occurs within the first 6 months of the medication [18, 28], the corresponding incidences from the Clarity AD and the US MDC cohort [11, 18, 28] are summarized in Table 3. In this study, 11 out of the 58 participants at risk for ARIA (6.9% for ARIA-E and 12.1% for ARIA-H) developed ARIA of varying severity. All four ARIA-E occurred before the 6-month follow-up, consistent with prior reports. The incidence of ARIA-E was lower than the 12.9% reported in a US memory clinic cohort [18] and below the 11.6% observed in the overall Clarity AD population [11, 28]. All ARIA cases in this study were detected through scheduled MRI monitoring and were asymptomatic. Treatment was temporarily suspended in three participants due to ARIA, with two resuming lecanemab therapy after the resolution of edema. These findings suggest that lecanemab is generally tolerated in the Chinese population, characterized by lower incidence and milder manifestations of ARIA. Consequently, lecanemab appears to have a more favorable safety profile in Asian populations, particularly in Chinese cohorts. This may be associated with the lower

Table 4 ARIA cases

	Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6	Patient 7	Patient 8	Patient 9	Patient10	Patient 11
Sex/age,y	F/76	F/69	F/75	M/74	M/57	F/79	M/69	M/67	M/60	M/69	M/69
Baseline MMSE	20	20	16	24	20	17	19	20	22	24	9
Education, y	9	6	16	16	6	12	16	16	16	9	0
ApoE	*ε3/ε3*	*ε4/ε4*	*ε3/ε4*	*ε3/ε4*	*ε3/ε4*	*ε3/ε4*	*ε2/ε3*	*ε2/ε3*	*ε3/ε3*	*ε3/ε4*	*ε4/ε4*
ARIA onset, months after the 1st infusion	3	3	2	6	3	3	6	12	3	2	2
ARIA-E	/	Moderate	Moderate	/	Mild	/	/	/	Moderate	/	/
ARIA-H (Number)	Moderate MCH (8)	/	Moderate MCH (5)	Moderate MCH (5)	/	Mild MCH (1)	Mild MCH (2)	Mild MCH (1)	/	Mild MCH (1)	Mild MCH (1)
ARIA locations	Bilateral occipital lobe	Bilateral occipital lobe	Bilateral occipital lobe	Bilateral posterior horns of the lateral ventricles	Right posterior horn of the lateral ventricle	Left temporal lobe	Brainstem, Adjacent to the left lateral ventricle	Right occipital lobe	Right occipital lobe	Right occipital lobe	Brainstem
MMSE (months after ARIA)	21 (6)	21 (6)	22 (3)	23 (3)	24 (3)	22 (3)	19 (6)	18(0)	27(0)	/	/
Baseline MCH	2	1	2	1	0	1	0	0	1	1	0
Baseline PVWMHs Fazekas	3	2	1	3	1	1	3	2	1	2	2
Baseline DWMHs Fazekas	3	2	1	3	2	1	2	2	1	2	2
Outcomes	Suspension	Suspension[a]	Suspension[b]	Dosage reduced to 400 mg with close monitoring	Continue medication with close monitoring	Continue medication with close monitoring	Continue medication with close monitoring	Continue medication with close monitoring	Suspension	Continue medication with close monitoring	Continue medication with close monitoring

Abbreviations MMSE Mini-Mental State Examination, *MCH* microhemorrhage, *PVWMH* periventricular white matter hyperintensities, *DWMH* deep white matter hyperintensities, *F* female, *M* male, *ARIA-E* amyloid-related imaging abnormalities with edema/effusion, *ARIA-H* amyloid-related imaging abnormalities with hemorrhage/hemosiderin deposits

[a]Medication was suspended because of moderate ARIA-E detected by MRI before the 7th infusion. Cerebral edema resolved after 3 months and the participant restarted medication with reduced dosage (400 mg). Mild ARIA-H was detected before the 10th infusion. The participant continued medication with close monitoring

[b]Medication was suspended because of moderate ARIA-E with moderate ARIA-H detected by MRI before the 5th infusion. The edema resolved after 5 months. The participant restarted medication after comprehensive assessments

frequency and severity of cerebral amyloid angiopathy (CAA) observed in Asian population [17, 30]. The fact that none of the participants were on anticoagulation therapy may be another contributing factor. Nevertheless, standardized MRI follow-up monitoring remains crucial. This facilitates timely clinical decisions regarding treatment modification or suspension.

It is known that *ApoE ε4* genotype is a major risk factor for ARIA, especially ARIA-E. ARIA-E and ARIA-H were more common in *ApoE ε4* carriers and were highest among *ApoE ε4* homozygotes within the lecanemab-treated group [6, 9, 10, 12, 22, 28, 29]. At treatment initiation, the higher risk of ARIA-E is also associated with higher dosage of and more microhemorrhages at baseline [1, 13, 20, 29]. In this study, among the ARIA-E cases identified, three were *ApoE ε4* carriers and one were a non-carrier. The absence of statistically significant association between ARIA-E and these risk factors,

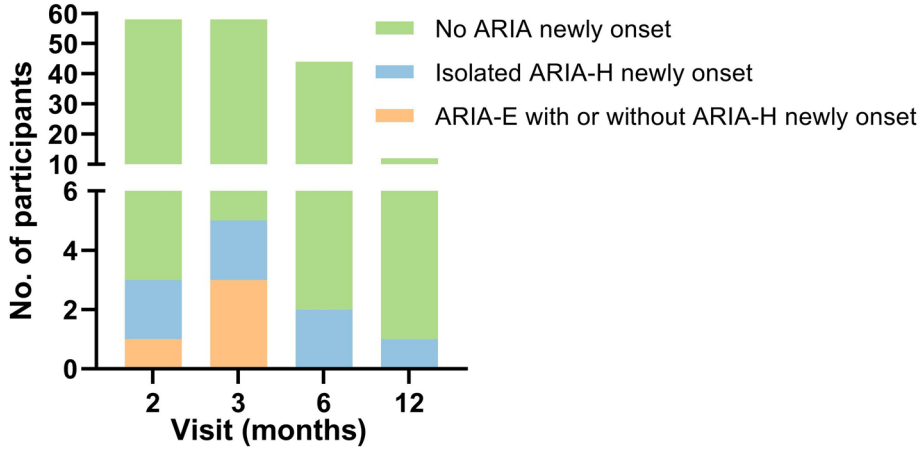

Fig. 3 Number of participants with or without newly onset ARIA. The analyses included only the 58 participants at risk for ARIA who received ≥ 4 infusions of lecanemab and underwent ≥ 1 monitoring MRI

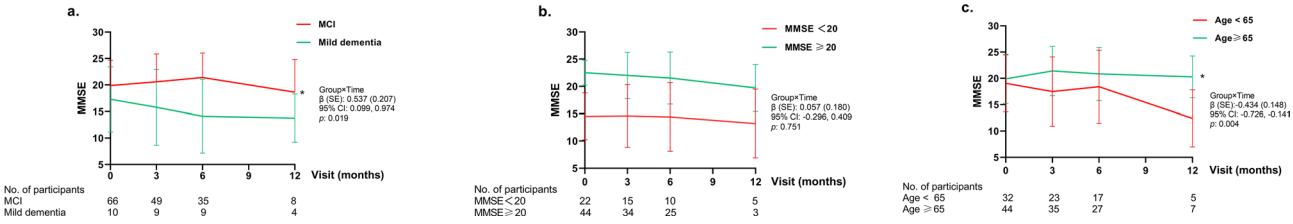

Fig. 4 Cognition assessments over 12-month follow-up. **a** Longitudinal MMSE score change between the MCI and mild-dementia groups. **b** Longitudinal MMSE score change in MCI participants, stratified by baseline MMSE < 20 versus MMSE ≥ 20. **c** Longitudinal MMSE score change between the EOAD and LOAD groups

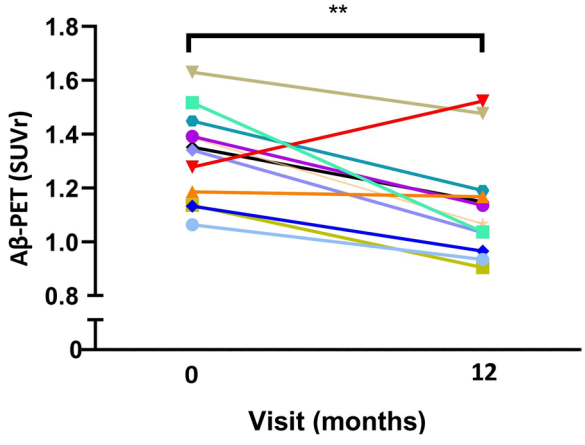

Fig. 5 Change in amyloid-β PET SUVr following 12 months of lecanemab therapy. The analysis included the 12 participants who received amyloid-β PET after 12 months of lecanemab therapy

particularly *ApoE ε4* genotype, may be related with the low event rate, which limited statistical power. Therefore, further investigations with larger sample size and longer-term follow-up are warranted to verify this relationship. Previous studies have shown a strong association between WMH and MCH, the latter representing the most common radiologic hallmark of ARIA-H [23]. In this study, participants with isolated ARIA-H exhibited higher baseline Fazekas scores for periventricular white matter hyperintensity, corroborating prior findings. However, considering the limited sample size, future studies with larger cohorts are needed to conduct more detailed analyses.

In this study, the development of ARIA-E or ARIA-H did not result in cognitive decline during the follow-up period, which may relate to the limited sample size and that all ARIA were mild to moderate and asymptomatic. Another case study even reported favorable long-term (54–65 months) cognitive outcomes following recurrent ARIA-E [33]. The limited sample size may constrain the generalizability of these findings. Therefore, further investigation with expanded cohorts is warranted to derive more robust conclusions.

Participants' cognitive progression during treatment was regularly assessed. The MCI group exhibited a significantly slower rate of decline in MMSE scores over time compared to the mild-dementia group. This finding suggests enhanced efficacy with earlier treatment, highlighting the importance of early detection, diagnosis, and initiation of therapy. The early-onset group experienced a significantly accelerated rate of MMSE decline over time. These results are consistent with the established findings [4, 15] and underscore the importance for timely intervention and closer monitoring in early-onset patients.

The MMSE score is closely associated with educational levels and culture differences, especially in Easten China, where a large proportion of the elderly population speaks different dialects. Therefore, the lecanemab therapy was initiated in patients based on their CDR-GS scores, following thorough communication with patients and their families. It was found that within the subgroup of participants with a baseline CDR-GS of 0.5 (MCI), no significant difference in the rate of MMSE decline over time between the MMSE < 20 and ≥ 20 groups. This finding suggests comparable efficacy of lecanemab in patients with lower baseline MMSE scores who nevertheless meet CDR-based MCI criteria. Consequently, for patients with lower education attainment, expanding eligibility criteria can be considered in clinical practice.

Consistent with the marked reduction in brain amyloid burden reported in the Clarity AD trial [28], a significantly decline in amyloid load was also observed after 12 months of lecanemab treatment, providing pathological support for the efficacy of lecanemab. However, one participant exhibited an increased amyloid burden after 12 months of treatment. She was a late-onset AD patient, *ApoE ε3/ε3*, who adhered fully to the weight-adjusted dose and completed 12 months of uninterrupted treatment without any IRSEs or ARIA. We suspected that this paradoxical response may be attributed to her innate immune profile, particularly microglial function, within the brain. It is recognized that microglial phagocytosis is essential for the efficacy of passive immunotherapy in AD. Impaired microglial phagocytosis in AD may compromise the Aβ-clearing effects of monoclonal antibodies [32]. Moreover, animal studies indicate that the brain's immune activation status can influence the efficacy of amyloid-targeted immunotherapy [14]. Thus, innate immune status, including microglial activation, may modulate therapeutic outcomes. Future studies incorporating microglia activation PET imaging could provide deeper insights.

Limitations

This study has several limitations. First, the small sample size may have contributed to false-negative results in some statistical analyses and reduced statistical power. The small number of participants with long-term follow-up markedly constrains our power to detect and analyze ARIA events, particularly with respect to ARIA-E risk factors. Similarly, the observed effects of ARIA-E and ARIA-H on cognition and function must also be interpreted cautiously. Larger-scale, real-world data are required. A more detailed statistical analysis will be conducted following future data collection efforts to expand the sample size. Second, the follow-up period for enrolled participants is relatively short, and efficacy and safety beyond that timeframe is unknown. Finally, the single-center design may introduce regional bias into the sample population.

Conclusions

This observational study provides insights into the application of lecanemab in a real-world clinical setting in Eastern China. The data support that lecanemab is generally tolerated in the Asian population. Adverse reactions, including infusion-related side effects and ARIA, were mild to moderate and less common than in the Clarity AD trial. In this limited dataset where subjects were treated according to the AUR recommendations – and where the ARIA-E cases were all mild or moderate and asymptomatic, ARIA-E does not appear to be associates with worsening cognition. Further investigation with respect to long-term cognitive outcomes and function in real-world clinical settings is warranted.

Abbreviations

AD	Alzheimer's disease
Aβ	Aggregated amyloid-β peptides
NFT	Neurofibrillary tangles
US	The United States
FDA	Food and drug administration
CDR-SB	Clinical dementia rating-sum of boxes
PET	Positron emission tomography
SUVr	Standardized uptake value ratio
MRI	Magnetic resonance imaging
ARIA	Amyloid-related imaging abnormality
ARIA-E	ARIA-effusion
ARIA-H	ARIA-hemorrhage
MCH	Micro-hemorrhage
ApoE	Apolipoprotein E
MCI	Mild cognitive impairment
MMSE	Mini-mental state global score
CDR-GS	Clinical dementia rating scale global score
IRSE	Infusion-related side effects
EOAD	Early-onset alzheimer's disease
LOAD	Late-onset alzheimer's disease
ADCs-ADL	Alzheimer's disease cooperative study-activity of daily life
NPI	Neuropsychiatric inventory
AURs	Appropriate use recommendations
FLAIR	Fluid attenuated inversion recovery
SWI	Susceptibility weighted imaging
DWI	Diffusion weighted imaging
PVWMH	Periventricular white matter hyperintensities
DWMH	Deep white matter hyperintensities
MTA	Medial temporal atrophy
LMMs	Linear mixed models
GLMMs	Generalized linear mixed models
CAA	Cerebral amyloid angiopathy

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13195-025-01886-5.

Supplementary Material 1.

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Authors' contributions
Yanxing Chen designed the study. Jia Shang, Siyan Zhong, Qingze Zeng, Shuai Zhao, Xinyi Zhang, Xiao Luo, Kaicheng Li, contributed to the data collection. Jia Shang, Li Shang, Siyan Zhong performed statistical analysis. Yanxing Chen, Jia Shang, Li Shang, Siyan Zhong contributed in analysis and interpretation of data. Yanxing Chen, Baorong Zhang, Zhirong Liu, Yaping Yan and Peiyu Huang participated in study supervision or coordination. Jia Shang, Li Shang, Siyan Zhong drafted the first version of the manuscript. All authors assisted in manuscript revision and finalization.

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Data availability
Raw anonymized data for statistical analysis are available upon reasonable request. All participants will be asked to sign a data access agreement.

Declarations

Ethical declaration and consent to participate
The study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. The protocol, consent form, and other relevant documentation were approved by the Human Research Ethics Committee of the Second Affiliated Hospital of Zhejiang University School of Medicine (Approval Number: 2024–1391). All participants voluntarily took part in the study and signed the informed consent form.

Consent for publication
All authors read and approved the final manuscript.

Competing interests
The authors declare no competing interests.

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